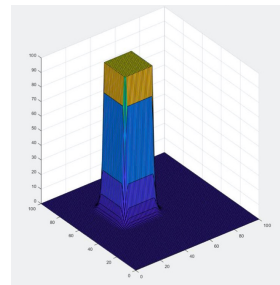
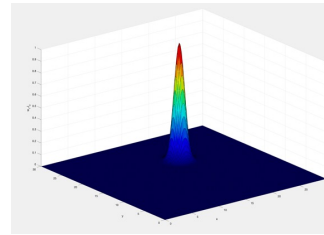


# HW#2 – Due Wed 10/16 at Midnight

In Canvas: Upload your code and a brief instruction on how to run it, together with a pdf document containing the data and plots requested below. Please upload a zip file containing all the files. The name of your files should follow this rule: *family name\_first name\_HW2*(e.g. Bezo\_Nicola\_HW2.zip).

1. Create a 100m x 100m workspace
2. Create an attractive potential field centered at  $q_{goal}=(80,20)$ ; Plot the potential field **(20pt)**
3. Create a square obstacle with the following coordinates  $\{(30,50)(50,50)(50,70)(30,70)\}$  and implement a radial repulsive potential field centered in the obstacle; Plot the potential field **(20pt)**
4. Combine the two potential fields and plot the resulting field **(10pt)**
5. Create a simple point-mass holonomic robot centered at  $q_{start}=(10,80)$  and use the resulting field to navigate the robot from  $q_{start}$  to  $q_{goal}$ .  $V_{max} = 3\text{m/s}$ . Plot the vehicle trajectory in the workspace and show the animation as the robot navigates to the goal **(30pt)**
6. Repeat 5. using the non-holonomic robot designed in HW#1 **(20pt)**
7. extra points: repeat 3-7 implementing a repulsive potential field that follows the shape of the obstacle **(10pt)**



## Useful Matlab commands for 3D plotting

- `surf()`
- `contour()`